

Parallel Operation Of Virtual Synchronous Machines With Frequency Droop Control

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Abstract—Studies had shown that the transient stability of a PV-Hydro micro-grid could be improved using a virtual synchronous machine (VSM), by injecting virtual inertia into the system. In the case of an interconnected system with multiple numbers of PV-Hydro micro-grid, a number of VSMs need to be operated in parallel. The parallel operated VSMs shall share the transient load in proportion to their respective capacities. Otherwise, the storage battery of one VSM will discharge more and other storage battery will be less discharged. This paper presents a control logic for parallel operated VSMs so that they share the transient load in proportion to the capacity of respective VSMs. Frequency droop control approach is used in the control logic to ensure proper load sharing. The control logic is simulated in MATLAB\ Simulink and results show that the frequency droop control approach is good enough for parallel operation of VSMs.

Index Terms—Energy storage, PV-Hydro micro-grid, Power system stability, Power electronic controller.

I. INTRODUCTION

Penetration of renewable energy sources like photovoltaic (PV) into an electric grid to meet the load demand and reduce dependency on non-renewable sources like fossil fuels, mineral oils, etc. is a matter of great interest. However, the high penetration of PV systems brings problems related to the stability of the system [1]. Micro-grids with a combination of small hydro generator and PV system has been developed to improve the reliability of PV in [2]. However, PV-hydro grids have a poor frequency stability owing to the decreased inertia of the system due to the interconnection of inertia-less PV systems. Thus, large changes in the load and small faults can lead to a drastic change in the system frequency with large a rate of change of frequency (ROCOF). Virtual Synchronous Machine (VSM) can be used to improve the frequency stability of micro-grids by emulating inertial and damping properties like that of a real synchronous machine through a properly controlled inverter [3-5].

There is a limit for the penetration level of PV systems into a micro-grid in order to ensure that the frequency of operation of the system remains within normal operating range. As per the standard for micro-grid, the maximum allowable frequency

deviation in steady state operation is $\pm 2\%$ of the nominal rated frequency and recommends that the frequency should not exceed $\pm 15\%$ of the nominal rated frequency during the transient period and the maximum recommended ROCOF value is $\pm 1\%$ of the rated system frequency [6-7]. The operation of generators outside the recommended levels can lead to degraded generation performance. Moreover, sustained periods of frequency variations affect proper operation of frequency dependent loads. A PV-hydro micro-grid system exhibits fast frequency dynamics, which cannot be handled by the conventional governors of hydro systems, responsible for the primary frequency control of the system. Large frequency deviations at high ROCOF thus may occur in such systems tripping frequency relays and causing unnecessary outages. Hence, added inertial response is needed to provide better dynamic frequency support during the transient periods of frequency deviations.

A VSM emulates virtual inertia by using some amount of stored energy controlled by a power electronics converter to absorb or inject power from/to the grid as necessary like the kinetic energy absorption or injection by a synchronous machine [7]. In an interconnected power system, where a number of PV-hydro micro-grids are present and each of them has its own VSM, it becomes necessary that the VSMs share the transient load in proportion to their respective capacities. Otherwise, the battery of larger VSM may get less discharged and the battery of smaller VSM may get more discharged. A coordinated operation of parallel operated VSMs is required to achieve proper transient load sharing. This paper presents a simulation study on operating parallel VSM units with proper transient load sharing through a frequency droop based controller.

The paper is organized as follows: Section II presents the basic concept of a VSM. Section III describes the benchmark used to analyze the parallel operation of VSMs. Section IV presents the simulation results performed in MATLAB/ Simulink. Section V discusses the conclusions of the paper and Section VI provides some recommendations for future work.

II. VIRTUAL SYNCHRONOUS MACHINE (VSM)

A VSM is a short-term energy storing device with power electronics circuit and a dispatching algorithm that makes it behave like a synchronous machine. VSMs emulate the inertial and damping properties of a synchronous machine and inject power into the grid or absorb the power from the grid depending upon whether the change in frequency is negative or positive, much like kinetic energy is injected or absorbed in synchronous machines. With these properties, a VSM can reduce the frequency variations during the transient periods.

Fig.1 shows PV-hydro microgrid with VSM to improve the frequency stability against the large changes in load and Fig.2 shows its control logic. The grid frequency is sensed using a phase locked loop (PLL) [8] which is then compared with the desired frequency to calculate frequency change and ROCOF. Selecting suitable values for the constants K_I and K_P the power to be exchanged by the VSM with the grid, P_{VSM} is calculated using the equation:

$$P_{VSM} = K_I \frac{d\Delta f}{dt} + K_P \Delta f \quad (1)$$

Where, K_I is emulated inertia constant and K_P is the emulated damping constant. The constants K_I and K_P can be calculated as:

$$K_I = \frac{P_{VSM_NOM}}{(\frac{d\Delta f}{dt})_{max}} \quad (2)$$

$$K_P = \frac{P_{VSM_NOM}}{(\Delta f)_{max}} \quad (3)$$

Where, P_{VSM_NOM} is the nominal power rating of the VSM, $(\Delta f)_{max}$ is the maximum allowable change in frequency in the system and $(\frac{d\Delta f}{dt})_{max}$ is the maximum allowable ROCOF. The rating of VSM should be chosen such that it can supply the deficit power or absorb the surplus during the worst possible transient condition which can be found out from the limiting cases for the change in frequency and ROCOF.

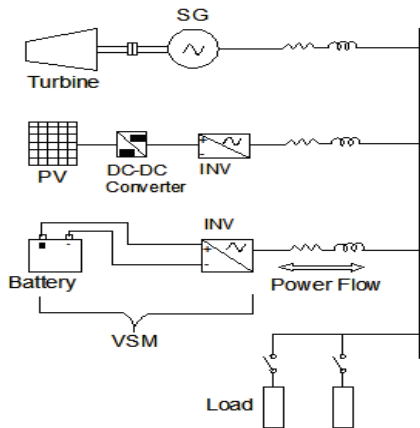


Fig. 1: A PV-hydro micro-grid with VSM to improve frequency stability.

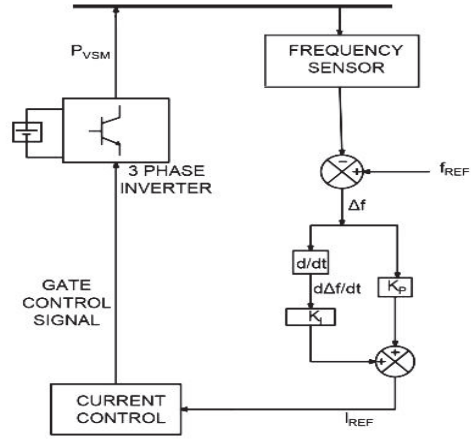


Fig. 2: Control logic of an implementation of VSM.

Based on the power calculated using (1), the current control block generates gate signals for inverter using the reference current set by the control algorithms which control the power to be injected into the grid or the power to be absorbed from the grid. Different types of current control techniques can be used to implement the current loop as described in [3].

III. PARALLEL OPERATION OF VSMs

In the case of microgrids covering a large area with multiple PV units, multiple numbers of VSMs have to be used and they ideally must share the transient load in proportion to their capacities. This paper deals with the coordinated control of two VSMs so that they share the transient load in proportion to their capacities. Fig.3 shows the benchmark that was developed for analyzing the parallel operation of two VSMs with their respective micro-grids.

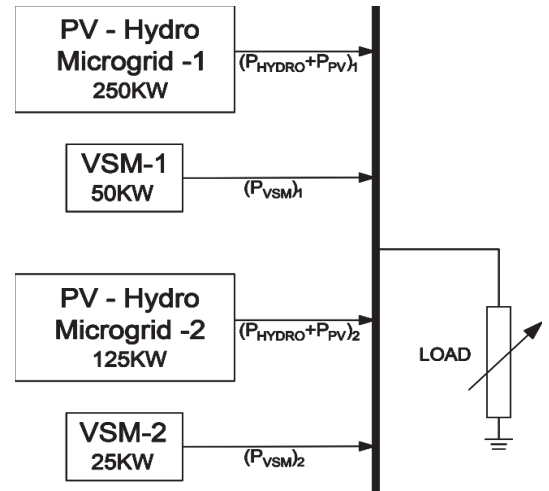


Fig. 3: Simulation benchmark used to study the parallel operation of two VSM units.

Two units of hydro generators of 250kW and 125 kW capacities are considered for the simulation study. Both hydro generators are synchronous generators driven by respective water turbines with automatic speed governor and respective excitation system controller. The nominal power rating of VSM of 250kW hydro generator is taken as 50 kW and that for 125 kW hydro generator is taken as 25 kW. These values are the maximum amount of power the VSM can inject during the transient period. The control logic of both VSMs is as shown in Fig.2 however, a frequency droop control logic needs to be incorporated so that they share the transient load in proportion to their capacities.

Fig.4 illustrates how frequency droop control can be implemented in parallel operated VSMs so that they share the transient load in proportion to their capacities. The slope m_1 of the frequency droop characteristic of larger VSM-1 is made less than the slope m_2 of the frequency droop characteristic of the smaller VSM-2 proportionately. It has been shown in [9] that droop controllers are mathematically equivalent to VSMs. Based on this concept, the damping constant K_P shown in Fig.2 is equal to reciprocal of the slope m . The value of K_P for the larger VSM was calculated by using (3) and the K_P for the smaller VSM was then calculated in proportion to its capacity. Similarly, the values of K_I for larger and smaller VSMs were calculated using (2).

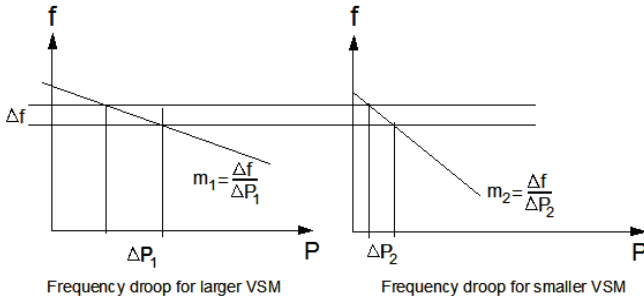


Fig.4: Frequency droop setting for parallel operation of multiple VSMs.

IV. SIMULATION RESULTS

In order to verify the proportionate load sharing by parallel operated VSMs, the benchmark developed in Fig.3 was implemented in MATLAB\Simulink along with the control logic. The maximum allowable frequency deviation and ROCOF were taken as $\pm 2\%$ and $\pm 1\%$ respectively during normal steady state operation. The maximum allowable frequency deviation was taken to be $\pm 15\%$ during the transient period as described above. Selecting a base power of 250kW and a base frequency of 50Hz to normalize the calculations in terms of per-unit (p.u.) system. The nominal power of larger VSM, P_{VSM_NOM} rated at 50KW is 0.2p.u. The following calculations were then performed to estimate the parameters for the larger VSM:

$$\left(\frac{d\Delta f}{dt}\right)_{max} = 1 \frac{Hz}{s} = 0.02 \text{ p.u.}$$

$$\Delta f_{max} = 2 \text{ Hz} = 0.04 \text{ p.u.}$$

$$K_{P1} = \frac{P_{VSM_NOM}}{\Delta f_{max}} = 5 \text{ p.u.}$$

$$K_{I1} = \frac{P_{VSM_NOM}}{\left(\frac{d\Delta f}{dt}\right)_{max}} = 10 \text{ p.u.}$$

Similar calculations for the smaller VSM unit rated at 25kW nominal power resulted in values of 2.5 p.u. and 5 p.u. for K_{P2} and K_{I2} respectively.

Fig.5 shows the simulation diagram of the parallel operation of two VSMs in MATLAB\Simulink. Fig.6 shows the details of the hydro generating unit, which consists of turbine governor model and excitation control model. Fig.7 shows the details of VSM model, developed as per the logic shown in Fig.2. Fig.8 shows the details of the hysteresis band current controlled inverter.

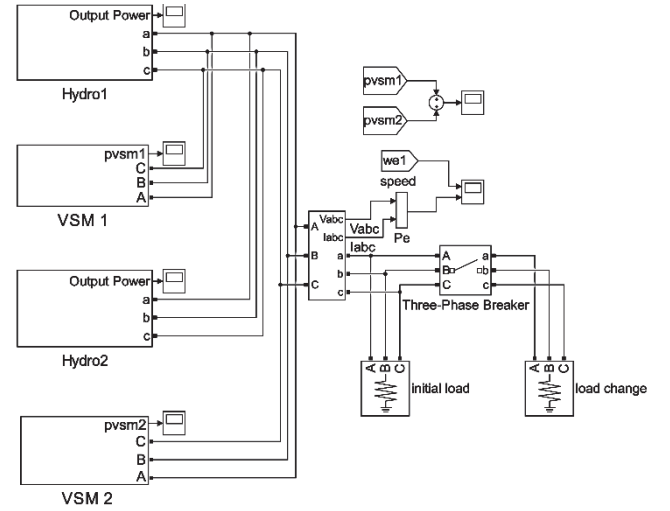


Fig.5: Simulation diagram of the parallel operation of two VSMs.

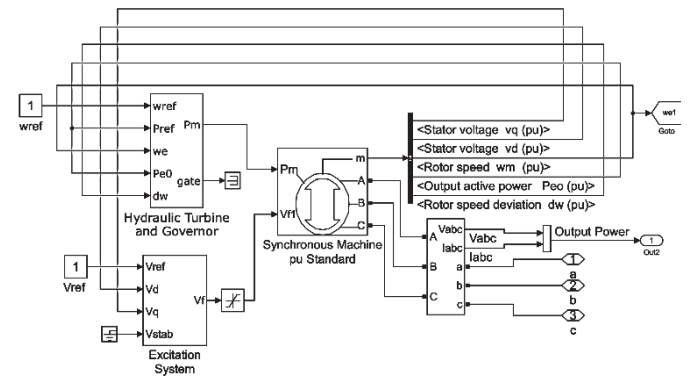


Fig.6: Details of a model of the hydro generating unit.

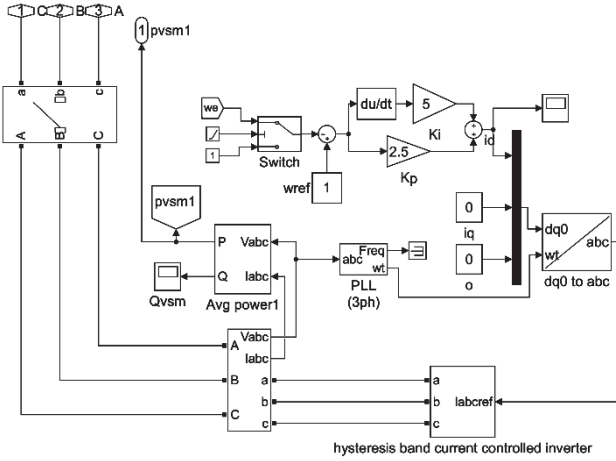


Fig. 7: Details of VSM block.

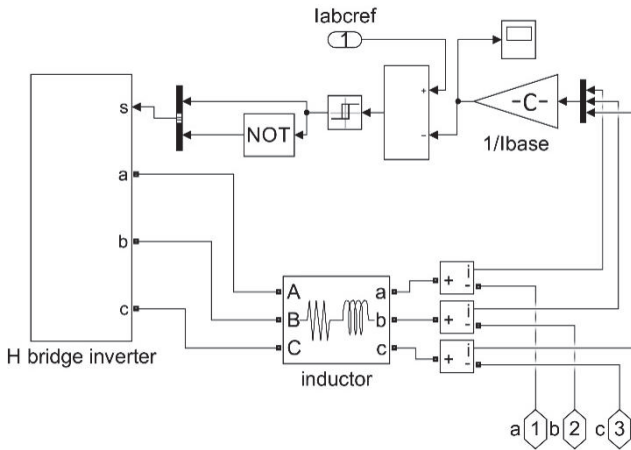


Fig. 8: Details of Hysteresis Band Current Controlled Inverter

The two hydro generators are operated in parallel to feed a common load of 100kW up to 10 s. At 10 s, a step load of 100 kW is switched on in parallel. The simulation is carried out for two different cases: without VSMs and with VSMs. The simulation results are shown in Fig.9.

Fig.9(a) shows the frequency response of the parallel operated hydro generators without VSM. From 0 to 10 s the system frequency is stable at 1 p.u. When a step load of 100 kW is switched ON at $t = 10$ s, the system frequency dips down to 0.725 p.u. and comes back to 1 p.u. after 35 s. In this case, the frequency has decreased below the recommended value of 0.85 p.u. Fig.9 (b) shows the frequency response of the parallel operated hydro-generators with VSMs. Again, from 0 to 10 s the system frequency is stable at 1 p.u. When the same step load of 100 kW, as in the previous case is switched ON at $t = 10$ s, the system frequency dips down to 0.92 p.u. only and comes back to 0.98 p.u. after 35 s. In this case, the frequency has not decreased below the acceptable value of 0.85 p.u. due to the

dynamic frequency support provided by the VSM units. Fig.9(c) shows the transient power delivered by VSM-1 and VSM-2 respectively. The larger VSM-1 has delivered 0.138 p.u. of power whereas, the smaller VSM-2 has delivered 0.068 p.u. of power which is in proportion to their capacities as expected.

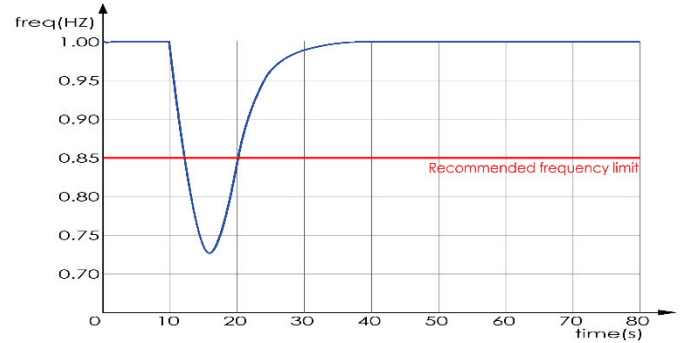


Fig.9(a): Frequency response of the parallel operated hydro generators without VSM

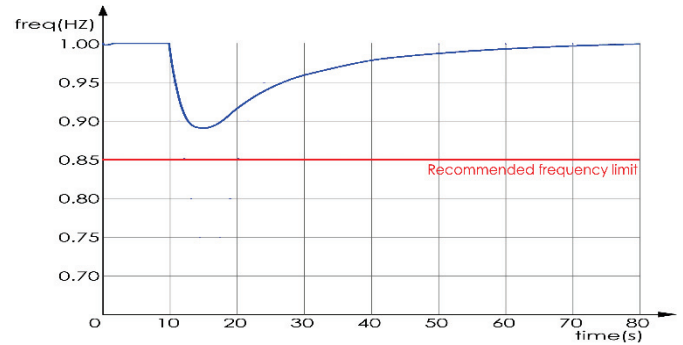


Fig.9(b): Frequency response of the parallel operated hydro generators with VSM

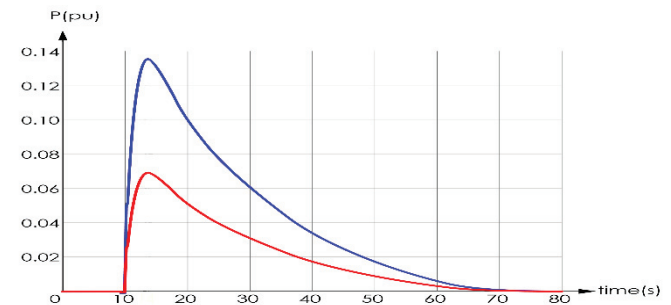


Fig. 9(c): Transient power shared by VSM-1 and VSM-2.

From the above results, it can be concluded that the VSMs help to improve the frequency stability in a PV-hydro microgrid during a large change in load and proportional power sharing of VSMs can be achieved by implementing a frequency droop controller.

V. CONCLUSIONS

In PV-hydro micro-grids, large load changes can cause significant changes in system frequency at a very high ROCOF which in some cases may lead to frequency going out of permissible limits. The low inertia of PV-hydro systems is responsible for these frequency variations. Additional inertia is required in the system to reduce these frequency variations. The deficit inertia in such autonomous PV-hydro micro-grid systems can be emulated by a VSM. When multiple numbers of VSM are used in parallel, proportional load sharing between VSMs can be obtained by implementing frequency droop controller.

VI. RECOMMENDATIONS FOR FUTURE WORK

When multiple numbers of VSMs are installed in different locations of a micro-grid, the impedance of the line in the network of micro-grid may affect the proportional load sharing. Furthermore, the inclusion of line impedance may affect the performance of PLLs especially in the case of weak micro-grids like PV-hydro system. It is recommended that these issues shall be taken into account in the future works.

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